



BHABHA UNIVERSITY

DEPARTMENT OF ELECTRONICS AND COMMUNICATION
M.TECH III SEM DIGITAL COMMUNICATION
SCHEME

MASTER OF TECHNOLOGY

DIGITAL COMMUNICATION III SEMESTER

S.No.	Subject Code	Subject Name	Periods per week				Maximum Marks (Practical Slot)		Total Marks		Practical Record/ Assignm ent/Quiz /Present ation	TOTAL MARKS
			L	T	P	CREDITS	End Sem. Exam.	Tests (Two)	Assign ments/ Quiz	End Sem. Practical /Viva		
1.	EC-301	Information Theory & Coding	3	1	-	4	70	20	10	-	-	100
2.	EC-302	Optical Instrumentation & Measurement	3	1	-	4	70	20	10	-	-	100
3.	EC-303	Seminar	-	-	4	4	-	-	-	-	100	100
4.	EC-304	Dissertation part 1	-	-	8	8	-	-	-	120	80	200
		Total	6	2	12	20	140	40	20	120	180	500



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SYLLABUS

EC – 301 INFORMATION THEORY & CODING

UNIT 1

Introduction to uncertainty, information, entropy and its properties, entropy of binary memory less source and its extension to discrete memory less source, coding theorem, data compression, prefix coding, HUFFMAN coding, Lempel-Ziv Coding

UNIT 2

Discrete memory less channels, Binary symmetric channel, mutual information & its properties, channel capacity, channel coding theorem, and its application to BSC, Shannon's theorem on channel capacity, capacity of channel of infinite bandwidth, Bandwidth signal to noise Trade off, Practical communication system in light of shannon's theorem, Fading Channel.

UNIT 3

Group and field of Binary system Galois field and its construction in $GF(2^m)$ and its basic properties, vector spaces and matrices in $GF(2)$, Linear Block Codes, Systematic codes, and its encoding circuits, syndrome and error detection, minimum distance, error detecting and correcting capabilities of block code, Decoding circuits, Probability of undetected error for linear block code in BSC, Hamming code and their applications.

UNIT 4

Cyclic codes and its basic properties, Generator & parity check matrix of cyclic codes, encoding & decoding circuits, syndrome computation & error detection, cyclic Hamming codes.

UNIT 5

Introduction to BCH codes, its encoding & decoding, error location & correction. Introduction to convolution codes, its construction & viterbi algorithm for maximum likelihood decoding.

REFERENCE BOOKS:

1. Digital Communication by Haykins Simon Wiley Publ.
2. Error control Coding: Theory and Application, by Shu Lin and Costello, PHI
3. Modern analog and Digital Communication system, by B.P. Lathi
4. Digital Communication by Sklar, Pearson Education
5. Principal of Communication system by Taub & Schilling, TMH
6. Error Correcting Codes by Peterson W., MIT Press



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SYLLABUS

EC – 302 OPTICAL INSTRUMENTATION & MEASUREMENT

UNIT 1

Optical Instrument: Optical Time Domain Reflector, Optical low Coherence Reflect meter, Optical Spectrum Analyzer Optical power and energy meter, Monochrometer, CCD, Ellipsometer, transducer, Lock in Amplifier, Box car Average.

UNIT 2

Fiber Optics Component and Devices: Direction Couplers, beam splitters, switches, modulations, connectors, couplers, polarizer, polarization controllers, amplifiers, fiber laser, reflector, wavelength filters, polarizing beam splitter, wavelength division multiplexes, fiber optic isolator etc.

UNIT 3

Fiber optic sensors: Pressure, temperature, strain, Magnetic & Electric field sensors based on characteristics like intensity, phase, polarization, frequency and wavelength of light wave

UNIT 4

Fiber optic Measurement: Introduction to measurement techniques

- i) Multimode Fiber: Refractive Index Profile, Geometric Measurement, Numerical Aperture, Total Attenuation, Scattering Loss and differential mode loss, Non destructive loss Measurement (OTDR), Transmission Bandwidth and dispersion, Bandwidth of Jointed fiber, Differential Mode Delay (DMD)
- ii) Single Mode Fiber: Attenuation, Refractive Index Profile (RIP), Mode Field Diameter, Equivalent step Index (EXI) Profile, Mode Cut off Wave length and the Single Mode operating regime, Dispersion, Birefringence Measurement, Measurement of the Propagation constant of fiber mode

REFERENCE BOOKS:

1. Optical Fiber Communication By S. Senior
2. Fiber Optics Measurement By A. Ghatak, M.R. Shenoy
3. Fundamental Of Fiber Optics in Telecommunication & Sensors Systems
4. Introduction to Fiber Optics By A. Ghatak and Tyagrajan
5. Optical Fiber Sensors system And Application By B. Culshaw